Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 11 · Part C: Electronics for Signal Processing Level 3

Track: Analog and Digital Modulation, Matched-Filter Reception, BER, Link Budgets, Doppler, Carrier Recovery

Prerequisites: Problems 04, 07, 08 · **Estimated effort:** 14–18 hours

Modulation and Telemetry Links: Getting Bits from a Satellite to the Ground

A CubeSat's signal arrives about 10^{-14} of a watt strong, 10 kHz off frequency and sweeping. Simulate AM, FM and BPSK, reproduce the theoretical BER curve of the matched-filter receiver, recover a carrier with a Costas loop, track a Doppler S-curve, and close a link budget with its margin.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/11_modulation_and_telemetry_links/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis

sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

A satellite telemetry link is a chain of signal-processing problems. The transmitter modulates data onto a carrier; the signal crosses 1000 km of vacuum losing 10^{14} in power; the receiver must find the carrier that Doppler shift has moved by up to 10 kHz, lock to its phase, filter optimally, and decide each bit against noise. Whether it works is summarised by one number, E_b/N_0, and by the margin of that number over what the modulation needs.

In this problem you implement the physical layer yourself: modulation and demodulation, the optimal receiver, carrier recovery and a link budget. The results link directly to ground-station design for student CubeSats operating in the amateur UHF band, and to the matched-filter theory of problem 19.

Theoretical Framework

1. Analog Modulation

Amplitude modulation $s(t) = [1 + m\,x(t)]\cos 2\pi f_c t$ (with $|m\,x| \le 1$) is demodulated by an envelope detector; frequency modulation $s(t) = \cos\!\big(2\pi f_c t + 2\pi k_f \int x\,\mathrm{d}t\big)$ by differentiating the phase. FM occupies, by Carson's rule, $B \approx 2(\Delta f + f_m)$ for peak deviation Δf and maximum message frequency f_m.

2. BPSK and the Matched Filter

In binary phase-shift keying a bit $b \in \{\pm 1\}$ is sent as $b\sqrt{2E_b/T_b}\cos 2\pi f_c t$ over a bit time T_b. With additive white Gaussian noise of one-sided density N_0 the optimal (matched-filter) receiver correlates with the pulse over one bit and decides by the sign, with bit-error rate

$$\mathrm{BER} = \mathrm{Q}\left(\sqrt{\frac{2E_b}{N_0}}\right) = \tfrac{1}{2}\,\mathrm{erfc}\sqrt{\frac{E_b}{N_0}}, \qquad \mathrm{Q}(x) = \frac{1}{\sqrt{2\pi}}\int_x^\infty \mathrm{e}^{-u^2/2}\mathrm{d}u. \tag{1}$$

A phase error θ in the recovered carrier reduces the useful amplitude by $\cos\theta$: $\mathrm{BER} = \mathrm{Q}\left(\sqrt{2E_b/N_0}\,\cos\theta\right)$.

3. Link Budget

With transmit power P_t, antenna gains G_t, G_r, free-space path loss $L_{\text{fs}} = 20\log_{10}(4\pi df/c)$, other losses L_o and receiver noise temperature T_s (all in dB):

$$P_r = P_t + G_t + G_r - L_{\text{fs}} - L_o, \qquad \frac{C}{N_0} = P_r - 10\log_{10}(k_{\text{B}}T_s), \qquad \frac{E_b}{N_0} = \frac{C}{N_0} - 10\log_{10} R_b. \quad (2)$$

The margin is E_b/N_0 minus the value required for the target BER (for uncoded BPSK about 9.6 dB for 10^{-5}).

4. Doppler Shift and Rate

A transmitter at range $r(t)$ produces $\Delta f = -f_0\,\dot{r}/c$. In the straight-line pass model $r = \sqrt{d_{\min}^2 + (vt)^2}$, so $\Delta f(t) = -f_0 v^2 t/(c\,r)$, with maximum rate of change at closest approach, $|\dot{\Delta f}|_{\max} = f_0 v^2/(c\,d_{\min})$.

5. Carrier Recovery

A Costas loop multiplies the received signal by the local carrier $\cos(\hat{\phi})$ and $\sin(\hat{\phi})$, forms the error $e = I \cdot Q$ (insensitive to the data sign), filters it with a proportional–integral loop filter and steers a numerically controlled oscillator. It locks at $\hat{\phi} = \phi$ or $\phi + \pi$; the π ambiguity is resolved by differential encoding.

Data and Simulation Specification

Analog tests	Message: 1 kHz tone; carrier 20 kHz; f_s = 200 kHz; AM index $m = 0.5$; FM deviation $\Delta f = 5$ kHz.
BPSK	Bit rate 1200 bit/s, 8 samples per bit, $E_b/N_0 = 0 \ldots 10$ dB, at least 10^6 bits near the low-BER points; rectangular and root-raised-cosine pulses.
Carrier recovery	Frequency offset $\Delta f = 50$ Hz, phase offset 1.0 rad, $E_b/N_0 = 10$ dB.
Doppler data	`sp_common.synth.satellite_doppler_pass(f0=437e6, d_min_km=700, v_rel_kms=7, duration_s=120, fs=20000, snr_db=20)`, returns complex IQ and the true Doppler curve.
Link parameters	$P_t = 1$ W, $G_t = 0$ dB, $G_r = 12$ dB, $f = 437$ MHz, slant range 1500 km, $L_o = 3$ dB, $T_s = 500$ K, $R_b = 9600$ bit/s.
Real data (optional)	A SatNOGS observation of a CubeSat beacon at 437 MHz (audio or waterfall data from the SatNOGS network and database), or your own capture with a low-cost SDR dongle; extract the Doppler curve and compare with a predicted pass.

Your Task

1. **AM and FM (Stage 2).** Generate AM and FM signals of the 1 kHz tone, demodulate AM with an envelope detector (Hilbert magnitude) and FM by phase differentiation. Measure the occupied bandwidth of the FM signal and compare with Carson's rule; plot the demodulation distortion against modulation index.
2. **BPSK BER curve.** Simulate BPSK with matched-filter detection in AWGN. Plot BER against E_b/N_0 from 0 to 10 dB (log scale) with Wilson confidence intervals and compare with Eq. (1). Use the correct noise variance for your sampling (derive it).
3. **Receiver comparison.** Compare the matched filter with (a) sampling a single point per bit

and (b) an integrate-and-dump with the wrong window length; express each loss as a dB penalty at BER $= 10^{-3}$.

4. **Carrier recovery.** Implement a Costas loop for the BPSK signal with the given frequency and phase offset. Plot the phase error and the loop-filter output, show lock acquisition time versus loop bandwidth, demonstrate the π ambiguity and fix it with differential encoding; quantify the BER penalty against the $\cos\theta$ prediction at several fixed phase errors.
5. **Doppler tracking.** From the synthetic pass estimate the Doppler curve by short-time FFT peak interpolation (choose the window length with the chirp-resolution rule of problem 17), then fit the straight-line model for $d_{\min}$, v and t_0 by nonlinear least squares. Report the fit errors against the truth and the maximum Doppler rate, which sets the bandwidth a tracking loop needs.
6. **Link budget.** Evaluate Eq. (2) for the given parameters, compute the margin against uncoded BPSK at 10^{-5} and plot the margin against slant range from 500 km to 2500 km and against bit rate. Find the maximum bit rate that retains a 6 dB margin.
7. **Real pass.** Analyse your SatNOGS or SDR data: find the carrier, extract the Doppler curve, locate the time of closest approach and compare with a prediction from the orbital elements (if you have them).

Self-check targets

BER of BPSK: 0.0786 at 0 dB, 0.0125 at 4 dB, 2.39×10^{-3} at 6 dB, 1.9×10^{-4} at 8 dB and $\approx 10^{-5}$ at 9.6 dB. The maximum Doppler rate of the default pass is $f_0 v^2/(c\, d_{\min}) \approx 102\,\mathrm{Hz/s}$ and $|\Delta f| \le 5.25\,\mathrm{kHz}$ within the $\pm 60\,\mathrm{s}$ window. The link budget gives $L_{\mathrm{fs}} = 148.8\,\mathrm{dB}$, $P_r = -139.8\,\mathrm{dBW}$, $C/N_0 = 61.8\,\mathrm{dB\,Hz}$, $E_b/N_0 \approx 22.0\,\mathrm{dB}$, a margin of about 12.4 dB over 9.6 dB.

Aerospace Application Link

You must specify the student ground station for a 1 W UHF CubeSat. Using your link-budget and BER results, decide the lowest-cost receive antenna gain and the highest bit rate that still keeps a 6 dB margin at 10° elevation (slant range about 1500 km for a 500 km orbit), state the Doppler pre-compensation range and rate your receiver must follow, and estimate how many bytes you receive in one 8 min pass.

Deliverables

D1 **Analog modulation:** AM and FM modulators and demodulators, Carson's-rule check.

D2 **BPSK BER:** the simulated curve with confidence intervals versus theory, and the receiver-comparison penalties.

D3 **Carrier recovery:** Costas-loop notebook with lock curves, ambiguity demonstration and BER-versus-phase-error check.

D4 **Doppler analysis:** the extracted curve, model fit with errors and the maximum-rate calculation.

D5 **Link budget:** the calculation, margin plots and the real-pass analysis.

D6 **Memo (300 words):** "Specification of the ground station for a UHF CubeSat." Cite at least two external sources.

Bonus (a) Add rate-1/2 convolutional coding with Viterbi decoding and measure the coding gain. (b) Implement symbol-timing recovery with a Gardner detector. (c) Decode an AX.25 frame

from a real recording.

Learning Resources

Digital Communications

- Sklar, B. *Digital Communications: Fundamentals and Applications* (BPSK, matched filter, E_b/N_0, link budgets).
- Proakis, J.G., Salehi, M. *Digital Communications* (optimum receivers, carrier synchronisation).
- Gardner, F.M. *Phaselock Techniques* (loop filters and noise bandwidth).

Space Links

- Larson, W.J., Wertz, J.R. (eds.) *Space Mission Analysis and Design*, communications chapter (link budgets, antenna gains).
- Maral, G., Bousquet, M. *Satellite Communications Systems* (propagation, Doppler, noise temperature).

Data and Tools

- SatNOGS network and database: `https://satnogs.org`, `https://db.satnogs.org`
- SciPy `scipy.signal` and `scipy.special.erfc`: `https://docs.scipy.org/doc/scipy/reference/`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.